

Clothing E-commerce Website: Project Documentation

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1. Overview

The project is a mobile-first online clothing store for the Indian market, with a customer storefront, an admin panel, online payments and cash on delivery. The reference is broncopolos.com, a D2C apparel store selling polos, tanks and T-shirts.

Goals

- A customer can find a product, pay and track the order on a phone without help.
- Staff can run catalog, stock, orders, offers and homepage content without a developer.
- Payments, refunds and COD are reconciled and auditable.

In scope for version 1: storefront, admin panel, payment gateway, COD, courier integration, SMS/WhatsApp/email notifications, coupons and offers, wishlist, bulk order enquiry form.

Out of scope for version 1: native mobile apps, multi-vendor marketplace, multi-currency, loyalty points, design voting (listed as a phase 2 item).

2. Reference UI (from the screenshots)

The three mobile screenshots define the look and the main browsing pattern: a sticky header, image-led tiles in a 2-column grid, and a slide-in drawer menu.

Screen	Elements observed	Requirement it implies
All screens	Yellow scrolling strip: "COD Available All Over India"	Admin-editable announcement bar
All screens	Header: menu icon, logo, offers icon, search, account (green dot when logged in), bag	Sticky header, login state indicator, cart count
Home	Auto-sliding hero banner with dots ("Rugby Polos", "Festive offer: buy any 2 for ₹1199")	Banner carousel managed from admin, each slide links to a collection

Screen	Elements observed	Requirement it implies
Home	2-column category tiles: Regular Polos, Designer Polos, Summer Tanks, Compression T-Shirts	Category tiles with image, label and link, order set in admin
Listing (Men – Tank Tops)	2-column product cards: photo, wishlist heart, brand, product name (truncated), sale price ₹449, struck-through MRP ₹999, red "55% OFF" badge, "100% Cotton" tag	Product card component; discount % computed from MRP and sale price; fabric tag as a product attribute
Drawer menu	Promo image, Shop By Category (Regular Polos, Designer Polos), Vote For Designs, Shop By Store (599 Store, 699 Store), Bulk Orders Query, Offers, Logout	Menu tree editable in admin; price-point stores; bulk enquiry form; offers page

URL pattern seen: `/shop/category/men/tank-tops/`, so categories are nested and slugs are readable.

Offer types visible: percentage off MRP, fixed price-point stores (everything at ₹599 or ₹699), and bundle pricing (any 2 for ₹1199). The offer engine in section 6 must support all three.

Use your own brand name, logo, photos and artwork. Several reference designs use licensed characters and logos, which need a licence to sell.

3. User roles

There are two customer roles and four staff roles; staff permissions are enforced on the server, not only hidden in the UI.

Role	Can do
Guest	Browse, search, add to bag, check pincode, place an order with phone OTP at checkout
Customer	Everything a guest can, plus wishlist, saved addresses, order history, tracking, cancel, return or exchange, reviews
Owner (super admin)	Everything, including staff accounts, payment and shipping settings, refunds

Role	Can do
Catalog manager	Products, categories, images, stock, size charts
Operations	Orders, shipping labels, returns, COD reconciliation, stock adjustments
Marketing and support	Banners, menu, announcement bar, coupons, offers, customer lookup, bulk enquiries; no refunds

Every admin write is recorded in an audit log with user, time, entity and before/after values.

4. Site map and navigation

The storefront has 5 core shopping pages, an account area and a set of static pages; the admin panel lives on a separate subdomain.

```

flowchart TD
    Home --> Listing[Category / store listing]
    Home --> Search
    Home --> Offers
    Listing --> Product[Product page]
    Search --> Product
    Product --> Bag
    Bag --> Checkout
    Checkout --> Confirm[Order confirmation]
    Confirm --> Track[Order tracking]
    Home --> Account
    Account --> Orders[My orders]
    Account --> Wishlist
    Account --> Addresses
  
```

The main purchase path runs down the middle: Home, listing, product, bag, checkout, confirmation.

Area	URL	Notes
Home	/	Announcement bar, banner carousel, category tiles, featured collections
Category listing	/shop/category/{parent}/{child}/	Filters, sort, infinite scroll or pagination

Area	URL	Notes
Price store	/store/599/ , /store/699/	Listing filtered to one fixed price
Product	/product/{slug}/	Variants, size chart, pincode check, reviews
Search	/search?q=	Suggestions while typing
Offers	/offers/	Active coupons and bundle deals
Bag and checkout	/bag/ , /checkout/	Checkout is a single page with 3 steps
Order tracking	/orders/{id}/	Also reachable from SMS link without login, protected by OTP
Account	/account/...	Orders, wishlist, addresses, profile
Bulk orders	/bulk-orders/	Enquiry form, goes to admin inbox
Static pages	/about-us/ , /contact/ , /shipping-policy/ , /returns/ , /privacy-policy/ , /terms-and-conditions/	Required for payment gateway approval
Admin	admin.yourdomain.com	Separate app, staff login with 2FA

Header (all pages): menu, logo, offers, search, account, bag with item count. **Drawer menu:** promo image, Shop By Category, Shop By Store, Bulk Orders Query, Offers, Login or Logout. Both are driven by a menu configuration stored in the database.

5. Customer-side requirements

Each requirement has an ID so it can be tracked in a task board and referenced in testing.

ID	Module	Requirement
C-01	Home	Announcement bar, banner carousel (auto-slide, swipe, dots), category tiles, all loaded from admin content

ID	Module	Requirement
C-02	Listing	2-column grid on mobile, 4 on desktop; card shows image, wishlist heart, brand, name, sale price, MRP, % off badge, fabric tag
C-03	Listing	Filters: size, colour, price, fabric, fit; sort: newest, price low to high, price high to low, discount
C-04	Listing	Out-of-stock products show a "Sold out" label and sort to the end
C-05	Search	Search by name, category, colour; suggestions after 2 characters; "no results" shows popular categories
C-06	Product	Image gallery with zoom, size and colour selectors, per-size stock state, size chart, fabric and care details, delivery estimate by pincode
C-07	Product	Add to bag, buy now, add to wishlist, share; related products; reviews with rating
C-08	Wishlist	Heart toggles from card and product page; guest hearts are kept locally and merged at login
C-09	Bag	Change size or quantity, remove, move to wishlist; shows MRP total, discount, coupon, shipping, payable amount
C-10	Offers	Apply coupon code; bundle offers (any 2 for ₹1199) apply automatically and show the saving; best offer wins when two conflict
C-11	Login	Phone number and OTP; OTP resend after 30 seconds; maximum 5 attempts per hour per number
C-12	Checkout	Step 1 address (pincode auto-fills city and state), step 2 payment method, step 3 review and pay
C-13	Checkout	Payment methods: UPI, cards, net banking, wallets, COD; prepaid discount and COD fee are configurable
C-14	Checkout	Stock is re-checked and reserved for 10 minutes when payment starts
C-15	Confirmation	Order number, items, amount, expected delivery; SMS, WhatsApp and email sent

ID	Module	Requirement
C-16	Orders	Order list, status timeline, courier tracking link, invoice PDF download
C-17	Orders	Cancel before dispatch in one tap; return or exchange request within 7 days of delivery with reason and photos
C-18	Refunds	Prepaid refunds go to the original payment method; COD refunds go to UPI ID or bank account collected in the return form; status visible to customer
C-19	Bulk orders	Form: name, phone, company, product, quantity, message; confirmation message on submit
C-20	Support	Contact page with phone, email, WhatsApp link; order-linked support request from the order page

Order status flow

```
stateDiagram-v2
    [*] --> Placed
    Placed --> Confirmed
    Placed --> Cancelled
    Confirmed --> Packed
    Confirmed --> Cancelled
    Packed --> Shipped
    Shipped --> Delivered
    Shipped --> RTO: refused or undelivered
    Delivered --> ReturnRequested
    ReturnRequested --> Returned
    Returned --> Refunded
    Cancelled --> Refunded: if prepaid
```

RTO means the courier returns the parcel to the warehouse; stock goes back in after a quality check.

C-16 to C-18 and C-20 deserve extra care. Public reviews of the reference store complain mostly about missing order updates, no way to cancel and unreachable support, so these features are where a new store earns trust.

6. Admin-side requirements

The admin panel is a desktop-first web app; everything a customer sees on the storefront is editable here without code changes.

ID	Module	Requirement
A-01	Dashboard	Today, 7-day and 30-day sales, order count, average order value, prepaid vs COD split, top 10 products, low-stock list
A-02	Products	Create and edit: name, slug, brand, description, MRP, sale price, fabric tag, fit, category, collections, images (drag to reorder), SEO title and description, active or draft
A-03	Variants	Size × colour matrix; each variant has SKU, stock, optional price override, barcode
A-04	Bulk tools	CSV import and export of products and stock; bulk price and status change
A-05	Categories	Nested tree (Men > Tank Tops), image, slug, sort order, show or hide
A-06	Price stores	Define a store by fixed price (599, 699); products join by tag or by price rule
A-07	Inventory	Stock per variant, low-stock threshold, manual adjustment with reason, stock history
A-08	Orders	List with filters (status, payment type, date, pincode); order detail with items, payments, shipment, notes, timeline
A-09	Orders	Confirm, pack, create shipment and print label and invoice, mark shipped, cancel, handle RTO
A-10	Returns	Approve or reject return requests, schedule reverse pickup, quality check result, restock, trigger refund
A-11	Payments	Transaction list, gateway status, refund initiation (owner only), COD remittance reconciliation against courier reports
A-12	Coupons	Code, percentage or flat, minimum bag value, max discount, per-user and total usage limits, start and end date, first-order only, category restrictions

ID	Module	Requirement
A-13	Offers	Bundle rule (buy N for ₹X from a collection), prepaid discount, COD fee, free-shipping threshold
A-14	Content	Announcement bar text, home banners (image, link, order, schedule), category tiles, drawer menu tree, drawer promo image, static pages
A-15	Customers	Search by phone or order number, order history, addresses, notes, block for COD abuse
A-16	Reviews	Approve, hide, reply
A-17	Bulk enquiries	Inbox with status (new, contacted, won, lost) and notes
A-18	Reports	Sales, GST (HSN-wise), returns and RTO rate, coupon usage; export to CSV
A-19	Staff	Staff accounts, roles from section 3, 2FA, audit log viewer
A-20	Settings	Store details, GSTIN, invoice series, payment gateway keys, courier keys, SMS and WhatsApp templates, serviceable pincodes

7. Payments, shipping and notifications

One payment gateway (Razorpay, Cashfree or PayU) covers UPI, cards, net banking and wallets; the gateway webhook, not the browser redirect, decides whether an order is paid.

```
sequenceDiagram
    participant U as Customer
    participant F as Storefront
    participant B as Backend
    participant G as Gateway
    U->>F: Tap Pay
    F->>B: Create order
    B->>B: Recalculate prices, reserve stock
    B->>G: Create gateway order
    G-->>B: Gateway order id
    B-->>F: Order id and amount
    F->>G: Open checkout, customer pays
    G-->>F: Payment id and signature
    F->>B: Verify payment
    B->>B: Check signature
    G->>B: Webhook payment.captured
    B->>B: Mark paid, confirm order
    B-->>U: SMS, WhatsApp, email
```

The backend never trusts an amount sent from the browser; it recalculates the bag from database prices before creating the gateway order.

Payment rules

- Webhooks are verified by signature and processed idempotently, so a repeated webhook cannot confirm an order twice.
- If payment is pending after 15 minutes, the order is marked failed and reserved stock is released.
- If the webhook reports success after the order was marked failed, the order is restored when stock allows, otherwise it is auto-refunded.
- No card data touches our servers; the gateway's hosted checkout keeps the site out of PCI scope.
- Refunds are raised from admin through the gateway API and their status is synced by webhook.

COD rules

- COD requires a verified phone OTP and a serviceable pincode.
- Configurable limits: maximum COD order value, COD fee, block list for customers with repeated refusals.
- Optional confirmation by WhatsApp or IVR before dispatch reduces RTO.

Shipping

- A courier aggregator (Shiprocket, Delhivery or similar) provides pincode serviceability, rate, AWB and label, tracking webhooks and reverse pickup.
- Tracking webhooks update the order status automatically: shipped, out for delivery, delivered, RTO.

Notifications

Event	SMS	WhatsApp	Email
OTP	Yes		
Order placed	Yes	Yes	Yes
Shipped, with tracking link	Yes	Yes	Yes
Out for delivery		Yes	
Delivered		Yes	Yes
Cancelled or refund initiated	Yes	Yes	Yes
Return approved and pickup scheduled		Yes	Yes
Abandoned bag reminder (phase 2)		Yes	

SMS in India needs DLT registration of sender ID and templates; WhatsApp needs approved templates from a Business API provider.

8. Data model

The product variant (one size and colour of one product) is the unit that carries SKU and stock, and every order line points to a variant.

```

flowchart LR
    Category --> Product
    Product --> Variant
    Product --> Image[Product image]
    Customer --> Address
    Customer --> Order
    Order --> OrderItem[Order item]
    OrderItem --> Variant
    Order --> Payment
    Order --> Shipment
    Order --> ReturnReq[Return request]
    Coupon --> Order
  
```

Arrows read as "has many", except Order item to Variant and Coupon to Order, which are references.

Entity	Key fields
customer	id, phone (unique), name, email, is_blocked_cod, created_at
address	id, customer_id, name, phone, line1, line2, city, state, pincode, is_default
category	id, parent_id, name, slug, image, sort_order, is_active
product	id, name, slug, brand, description, mrp, sale_price, fabric_tag, fit, category_id, hsn_code, gst_rate, status, seo_title, seo_description
product_image	id, product_id, url, alt, sort_order, colour
variant	id, product_id, sku (unique), size, colour, stock, reserved, price_override
collection, collection_product	Groups for price stores, bundles and home sections
wishlist_item	customer_id, product_id
cart, cart_item	cart id (guest token or customer_id), variant_id, qty
coupon	code, type, value, min_bag, max_discount, usage_limit, per_user_limit, starts_at, ends_at, rules
offer_rule	type (bundle, prepaid, shipping), config JSON, priority, active window
order	id, order_no, customer_id, address snapshot, subtotal, discount, shipping, cod_fee, total, payment_method, payment_status, status, coupon_code, placed_at
order_item	order_id, variant_id, name and price snapshot, qty, gst_amount
payment	order_id, gateway, gateway_order_id, gateway_payment_id, amount, status, raw webhook payload

Entity	Key fields
refund	payment_id or order_id, amount, method, status, gateway_refund_id
shipment	order_id, courier, awb, label_url, status, tracking events
return_request	order_item_id, type (return or exchange), reason, photos, status, qc_result
review	product_id, customer_id, rating, text, status
banner, menu_item, page, setting	Storefront content managed from admin
bulk_enquiry	name, phone, company, product, qty, message, status
staff_user, role, audit_log	Admin access and change history
stock_movement	variant_id, change, reason, reference, staff_user_id

Orders store snapshots of address, product name and price, so later edits to a product never change a past invoice. Money is stored as integers in paise.

9. API outline

A single REST API under `/api/v1` serves both apps; storefront routes use a customer token, `/admin` routes use a staff token with role checks, and `/webhooks` routes verify a provider signature.

Group	Endpoints
Auth	POST <code>/auth/otp/send</code> , POST <code>/auth/otp/verify</code> , POST <code>/auth/logout</code> , GET <code>/me</code>
Content	GET <code>/content/home</code> , GET <code>/content/menu</code> , GET <code>/pages/{slug}</code>
Catalog	GET <code>/categories</code> , GET <code>/products?category=&size=&colour=&price=&sort=&page=</code> , GET <code>/products/{slug}</code> , GET <code>/search/suggest?q=</code>
Pincode	GET <code>/pincode/{code}</code> returns serviceability, COD availability, delivery estimate

Group	Endpoints
Wishlist	GET /wishlist, PUT /wishlist/{productId}, DELETE /wishlist/{productId}
Cart	GET /cart, POST /cart/items, PATCH /cart/items/{id}, DELETE /cart/items/{id}, POST /cart/coupon, DELETE /cart/coupon
Checkout	POST /checkout/orders creates the order and the gateway order, POST /checkout/verify checks the payment signature
Orders	GET /orders, GET /orders/{id}, POST /orders/{id}/cancel, POST /orders/{id}/returns, GET /orders/{id}/invoice
Other	POST /reviews, POST /bulk-enquiries, POST /support-requests
Webhooks	POST /webhooks/payment, POST /webhooks/shipping
Admin	/admin/products, /admin/variants, /admin/categories, /admin/inventory, /admin/orders, /admin/returns, /admin/payments, /admin/coupons, /admin/offers, /admin/content, /admin/customers, /admin/reports, /admin/staff, /admin/settings with standard list, create, update, delete

Conventions: JSON, cursor or page pagination, consistent error shape `{code, message, fields}`, an `Idempotency-Key` header on order creation, and OpenAPI documentation generated from the code.

10. Tech stack and architecture

Recommended: a Next.js storefront, a React admin app and one Node.js API on PostgreSQL, with third-party services for payments, shipping, messaging and images.

flowchart LR

```

Shopper[Customer browser] --> Store[Storefront<br/>Next.js]
Staff[Staff browser] --> Admin[Admin panel<br/>React]
Store --> API[Backend API<br/>Node.js]
Admin --> API
API --> DB[(PostgreSQL)]
API --> Cache[(Redis)]
API --> Media[Image storage<br/>and CDN]
API --> PG[Payment gateway]
API --> Ship[Courier aggregator]
API --> Msg[SMS, WhatsApp,<br/>email]

```

Both apps talk only to the API; the API is the only component that holds gateway and courier credentials.

Layer	Choice	Why
Storefront	Next.js, TypeScript, Tailwind CSS	Server rendering for SEO and fast first load on mobile networks
Admin panel	React, TypeScript, a component library (shadcn/ui or Ant Design), TanStack Table	Data-heavy screens built quickly
Backend	Node.js with NestJS, Prisma ORM	One language across the stack, typed models
Database	PostgreSQL	Transactions for stock and orders
Cache and jobs	Redis with BullMQ	OTP limits, stock reservation expiry, notification and webhook queues
Search	PostgreSQL full-text first, Meilisearch when the catalog passes about 2,000 products	Avoids early complexity
Media	S3 or Cloudinary behind a CDN, WebP output, multiple sizes	Product photos are most of the page weight
Hosting	Vercel for storefront, AWS Mumbai or Railway for API and database	Low latency for Indian users
Monitoring	Sentry, uptime checks, GA4, Meta Pixel	Errors and marketing attribution
CI/CD	GitHub Actions, staging and production environments	Safe releases

Faster alternative: Medusa.js (open source, Node.js) already provides products, variants, carts, orders, discounts and an admin panel. Building the storefront and the India-specific integrations on top of it cuts roughly 6 to 8 weeks from the timeline in section 12.

11. Non-functional requirements

Targets are set for a mid-range Android phone on a 4G connection, since that is the typical customer device.

Area	Requirement
Performance	Largest Contentful Paint under 2.5 s on home, listing and product pages; images lazy-loaded and served as WebP; API responses under 300 ms at the 95th percentile
Capacity	500 concurrent shoppers and 50 orders per minute during a sale without errors; load-tested before launch
Availability	99.9% monthly uptime; daily database backups kept 30 days; restore tested once before launch
Security	HTTPS everywhere; server-side price and stock validation; OTP and login rate limits; admin 2FA; role checks on every admin route; secrets in environment variables; webhook signature checks; dependency scanning in CI
Privacy	Collect only name, phone, email, address; privacy policy and consent in line with India's DPDP Act; customer can request account deletion
SEO	Readable URLs, meta tags per product and category, sitemap.xml, robots.txt, Product and Breadcrumb structured data, canonical URLs on filtered listings
Accessibility	Alt text on product images, 44 px minimum tap targets, visible focus states, colour contrast at WCAG AA
Browser support	Last 2 versions of Chrome, Safari, Samsung Internet, Edge, Firefox

Area	Requirement
Legal and tax	GST-compliant invoice with GSTIN, HSN code and tax split; MRP display; published return, shipping, privacy and terms pages; grievance officer contact as required by the Consumer Protection (E-Commerce) Rules
Testing	Unit tests on pricing, offers and stock logic; end-to-end tests for the purchase path; gateway sandbox tests for success, failure, late webhook and refund

12. Milestones and timeline

A custom build takes about 16 weeks with 1 designer, 2 developers and part-time QA; phases 3 and 4 run in parallel.

Phase	Weeks	Deliverables
1. Discovery and design	1-3	Final requirement list, Figma designs for mobile and desktop, database schema, API contract
2. Backend foundation	3-6	Auth with OTP, catalog, variants, inventory, cart, offers engine, order APIs
3. Storefront	5-10	Home, listing, product, search, wishlist, bag, checkout, account, static pages
4. Admin panel	6-11	Modules A-01 to A-20
5. Integrations	10-13	Payment gateway, courier aggregator, SMS, WhatsApp, email, invoice PDF
6. Testing and hardening	13-15	End-to-end tests, payment edge cases, load test, security review, content upload
7. Launch	16	Production deploy, live test orders (prepaid and COD), monitoring, soft launch to a small audience

Start in week 1, in parallel with design: GST registration, current account, payment gateway KYC, courier aggregator account, DLT registration for SMS, WhatsApp Business API approval. Each takes 1 to 3 weeks and blocks phase 5.

Phase 2 backlog after launch: Vote For Designs, abandoned-bag reminders, photo reviews, loyalty points, recommendations, mobile app.

13. Open questions

These 8 answers change scope, cost or timeline and should be settled before phase 1 ends.

- ☐ Brand name, domain and logo for the new store?
- ☐ Custom build as documented, or Medusa.js or Shopify to launch faster?
- ☐ Who builds it: in-house developers, freelancers or an agency?
- ☐ How many products and variants at launch, and are product photos ready?
- ☐ Own warehouse and stock, or print-on-demand through a supplier?
- ☐ Which payment gateway and courier aggregator, and is the business KYC (GST, current account) in place?
- ☐ Return policy: 7 days, return and exchange both, or exchange only?
- ☐ Is Vote For Designs wanted in version 1 or later?